Controlling $\Delta^o_{k,j}$. For $t = t_o$, noting that $z_{t_o} = y$, we have

$$\begin{aligned}\Delta^o_{k,j} &= \mathbb{E}\left[\mathbb{1}\{y=k\}(x_{t_o,1} - \bar{x}_1)|y=k, q=j\right] - \mathbb{E}\left[\mathbb{1}\{y=k\}(x_{t_o,1} - \bar{x}_1)|q=j\right] \\ &= \left(1 - \frac{1}{N}\right)\mathbb{E}\left[x_{t_o,1} - \bar{x}_1|y=k, q=j\right] \\ &= \frac{N-1}{N}\left(\bar{p} + \sum_{i=1}^N a_{k,j,i}\Phi_1 w_E(i)\right),\end{aligned}$$

with $a_{k,j,i} \approx (\Phi_1 w_E(i))^\top \mathbb{E}\left[x_{t_o,1} - \bar{x}_1|y=k, q=j\right]$ thanks to near-orthonormality, and

$$\bar{p} = \mathbb{E}_{t_o}\left[\frac{1}{t_o}\sum_{s=1}^{t_o} p_s - \frac{1}{T}\sum_{t=1}^T \frac{1}{t}\sum_{s=1}^t p_s\right]$$

is a spurious positional mixture.

We then distinguish the following cases:

- If $j \neq k$ and $i = j$, since the trigger token j only appears at positions $t_o - 1$ and T, we have

$$a_{k,j,i} \approx \mathbb{E}_{t_o}\left[\frac{1}{t_o} - \frac{1}{T}\sum_{t=t_o-1}^T \frac{1}{t} - \frac{1}{T^2}\right] =: \gamma_T.$$

  We may expect t_o to be concentrated around $T/2$, in which case $\gamma_T \gtrsim \frac{2}{T} - \frac{1}{T} - \frac{1}{T^2} \geq \frac{C}{T} > 0$ for T larger than a small constant.
- If $j = k = i$, the two occurrences of the trigger happen one after the other, so it must be that $t_o = T$. Then

$$a_{k,j,i} \approx \frac{2}{T} - \frac{1}{T(T-1)} - \frac{2}{T^2} = \frac{2}{T} + O\left(\frac{1}{T^2}\right),$$

  for T larger than a small constant.
- If $i \neq j = k$, all tokens up to position $t_o - 2 = T - 2$ are i.i.d. uniform in $[N] \setminus \{j\}$, so that

$$a_{k,j,i} \approx \frac{T-2}{T(N-1)} - \frac{1}{T}\left((T-2)\cdot\frac{1}{N-1} + \frac{T-2}{(T-1)(N-1)} + \frac{T-2}{T(N-1)}\right) = O\left(\frac{1}{N}\right)$$

- If $i \neq j$ and $j \neq k$, all tokens except at positions $t_o - 1$, t_o and T (we have $t_o < T$) are uniform in $[N] \setminus \{j\}$. The triggers do not contribute anything to $a_{k,j,i}$ since $i \neq j$, and the output token may be also randomized by taking the average over $k \in [N] \setminus \{j\}$. We thus obtain

$$\frac{1}{N-1}\sum_{k \neq j} a_{k,j,i} \approx O\left(\frac{1}{N}\right).$$

In summary, we obtain

$$\frac{1}{N}\sum_{k=1}^N a_{k,j,i} \approx \begin{cases} O\left(\frac{1}{N}\right), & \text{if } i \neq j \\ \Omega\left(\frac{1}{T}\right), & \text{if } i = j. \end{cases}$$

Thus, when N is large, while T is moderate, the above sum leads to more signal in the $i = j$ terms compared to $i \neq j$. In particular, this yields

$$(\Phi_1 w_E(i))^\top \left(\frac{1}{N}\sum_{k=1}^N \Delta^o_{k,j}\right) \approx \begin{cases} O\left(\frac{1}{N}\right), & \text{if } i \neq j \\ \Omega\left(\frac{1}{T}\right), & \text{if } i = j, \end{cases}$$

so that this component in (24) acts precisely like the desired associative memory in (7).

It remains to show that the other components are negligible compared to this. It then suffices to show:

$$\frac{1}{N}\sum_{k=1}^N (\Delta^q_{k,j} + \Delta^r_{k,j}) \approx o\left(\frac{1}{T}\right).$$

Controlling $\Delta^q_{k,j}$. For $t \in \mathcal{T}_q$, note that we always have $z_t = j$ in the expectations, so that $\Delta^q_{k,j} = 0$ unless $k = j$. For $k = j$, we have $\Delta^q_{k,j} = O(1)$, so that

$$\frac{1}{N}\sum_{k=1}^{N} \Delta^q_{k,j} = O\left(\frac{1}{N}\right).$$

Controlling $\Delta^r_{k,j}$. Using that $\|x_{t,1} - \bar{x}_1\| \le C = O(1)$ for all t, we provide the following crude bound via the triangle inequality and Hölder inequality:

$$\Delta^r_{k,j} = \mathbb{E}\left[\sum_{t\in\mathcal{T}_r} \mathbb{1}\{z_t = k\}(x_{t,1} - \bar{x}_1) | y = k, q = j\right] - \mathbb{E}\left[\sum_{t\in\mathcal{T}_r} \mathbb{1}\{z_t = k\}(x_{t,1} - \bar{x}_1) | q = j\right]$$

$$\|\Delta^r_{k,j}\| \le C\left(\mathbb{E}\left[\sum_{t\in\mathcal{T}_r} \mathbb{1}\{z_t = k\} | y = k, q = j\right] + \mathbb{E}\left[\sum_{t\in\mathcal{T}_r} \mathbb{1}\{z_t = k\} | q = j\right]\right) \le \frac{2CT}{N},$$

since z_t is independent of y given $t \in \mathcal{T}_r$ and thus is uniform in $[N] \setminus \{j\}$, and $|\mathcal{T}_r| \le T$. We note, however, that $\Delta^r_{k,j}$ may be controlled much more finely by leveraging the similarities between the distributions of $z_t, t \in \mathcal{T}_r$ with or without conditioning on y.

Overall, we have shown that up to some spurious positional embeddings, W^2_K behaves as the desired associative memory from (7) when N is large enough, satisfying:

$$(\Phi_1 w_E(i))^\top W^2_K w_E(j) \approx \frac{\alpha\eta}{TN}\left\{\Omega\left(\frac{1}{T}\right)\mathbb{1}\{i = j\} + O\left(\frac{T}{N}\right)\right\} \tag{25}$$

We note that one may then amplify the gap between correct and incorrect associations by having a large enough step-size, which then makes the softmax more peaked and hence the attention more sparse and focused on correct associations.

B.3.3 Learning W^1_K

We now assume that W^2_O and W^2_K have learned the correct associations, and consider one gradient step away from zero-initialization on W^1_K. Note that when $W^1_K = 0$, the predictions of the model are still often near random chance. Indeed, the second layer attention will attend to all tokens starting at the first occurrence of the trigger, since all such tokens contain $\Phi_1 w_E(q)$ in their average attention, which activates the second-layer attention head. Then the output is likely to predict the trigger itself, which will be an incorrect prediction most of the time.

We may thus consider $\hat{p}(k|X) = 1/N$ at this stage as well. We also consider a simplified architecture where the first layer attention only uses positional embeddings in the key-query matrix, and only token embeddings in the value-output matrix. In particular, we have $x_t = w_E(z_t)$. Lemma 5 then gives the following form for W^1_K after one gradient step of step-size η:

$$\begin{aligned}
W^1_K = &\frac{\eta}{N}\sum_{k=1}^{N}\mathbb{E}_X\left[\frac{1}{T}\sum_{t=1}^{T} w_U(k)^\top \Phi_2 x_t \cdot \frac{1}{t}\sum_{s=1}^{t}(\Phi_1 x_s)^\top W^2_K x_T (p_s - \bar{p}_{1:t})p_t^\top | y = k\right] \\
&- \frac{\eta}{N}\sum_{k=1}^{N}\mathbb{E}_X\left[\frac{1}{T}\sum_{t=1}^{T} w_U(k)^\top \Phi_2 x_t \cdot \frac{1}{t}\sum_{s=1}^{t}(\Phi_1 x_s)^\top W^2_K x_T (p_s - \bar{p}_{1:t})p_t^\top\right] \\
&- \frac{\eta}{N}\sum_{k=1}^{N}\mathbb{E}_X\left[w_U(k)^\top \Phi_2 \bar{x}_{1:T} \cdot \frac{1}{T}\sum_{t=1}^{T}\frac{1}{t}\sum_{s=1}^{t}(\Phi_1 x_s)^\top W^2_K x_T (p_s - \bar{p}_{1:t})p_t^\top | y = k\right] \\
&+ \frac{\eta}{N}\sum_{k=1}^{N}\mathbb{E}_X\left[w_U(k)^\top \Phi_2 \bar{x}_{1:T} \cdot \frac{1}{T}\sum_{t=1}^{T}\frac{1}{t}\sum_{s=1}^{t}(\Phi_1 x_s)^\top W^2_K x_T (p_s - \bar{p}_{1:t})p_t^\top\right].
\end{aligned}$$

Note that since W^2_O and W^2_K already captured the desired associations at this stage, we have

$$w_U(k)^\top \Phi_2 x_t \approx \alpha\, \mathbb{1}\{z_t = k\} \quad \text{and} \quad (\Phi_1 x_s)^\top W^2_K x_T \approx \alpha'\, \mathbb{1}\{z_s = z_T\},$$

for some $\alpha, \alpha' > 0$. Recall that in our model, we have $z_T = q$ with probability one (q is the trigger token), and that q only appears twice: once at position $t_q := t_o - 1 < T$ and once at position T. We then have, for any $t > 1$,

$$W_K^1 p_t \approx \frac{\eta\alpha\alpha'}{NTt} \sum_{k=1}^{N} (A_{t,k} - B_{t,k} - C_{t,k} + D_{t,k}),$$

with

$$A_{t,k} = \mathbb{E}[\mathbb{1}\{z_t = k\}\, \mathbb{1}\{t_q \le t\}(p_{t_q} - \bar{p}_{1:t}) | y = k] \tag{26}$$

$$B_{t,k} = \mathbb{E}[\mathbb{1}\{z_t = k\}\, \mathbb{1}\{t_q \le t\}(p_{t_q} - \bar{p}_{1:t})] \tag{27}$$

$$C_{t,k} = \mathbb{E}[r_k\, \mathbb{1}\{t_q \le t\}(p_{t_q} - \bar{p}_{1:t}) | y = k] \tag{28}$$

$$D_{t,k} = \mathbb{E}[r_k\, \mathbb{1}\{t_q \le t\}(p_{t_q} - \bar{p}_{1:t})], \tag{29}$$

where $r_k := \frac{1}{T}\sum_{t=1}^{T} \mathbb{1}\{z_t = k\}$. We have

$$\begin{aligned}
A_{t,k} &= \mathbb{E}[\mathbb{1}\{z_t = k\}(\mathbb{1}\{t_q = t-1\} + \mathbb{1}\{t_q \in [t-2] \cup \{t\}\})(p_{t_q} - \bar{p}_{1:t}) | y = k] \\
&= \mathbb{P}(t_q = t-1 | y = k)(p_{t-1} - \bar{p}_{1:t}) + \frac{1}{N} \sum_{s \in [t-2] \cup \{t\}} \mathbb{P}(t_q = s | y = k)(p_s - \bar{p}_{1:t}) \\
&= \mathbb{P}(t_q = t-1)(p_{t-1} - \bar{p}_{1:t}) + \frac{1}{N} \sum_{s \in [t-2] \cup \{t\}} \mathbb{P}(t_q = s)(p_s - \bar{p}_{1:t}) \\
&= \mathbb{P}(t_q = t-1)(p_{t-1} - \bar{p}_{1:t}) + O\left(\frac{1}{N}\right),
\end{aligned}$$

since the distribution of t_q is the same regardless of y. We proceed similarly for the other quantities and obtain the following:

$$\begin{aligned}
B_{t,k} &= O\left(\frac{1}{N}\right) \\
C_{t,k} &= \frac{\mathbb{P}(t_q = t-1)}{T}(p_{t-1} - \bar{p}_{1:t}) + O\left(\frac{1}{N}\right) \\
D_{t,k} &= O\left(\frac{1}{N}\right).
\end{aligned}$$

This yields the following associative memory behavior, for $t > 1$:

$$p_s^\top W_K^1 p_t \approx \frac{\eta\alpha\alpha'(T-1)}{T^2 t} \left\{ \mathbb{P}(t_q = t-1) \left(\mathbb{1}\{s = t-1\} - \frac{1}{t}\, \mathbb{1}\{s \in [t]\} \right) + O\left(\frac{1}{N}\right) \right\},$$

which matches the desired "previous token head" behavior from (7) when N is large. As in the case of W_K^2, we may then "saturate" the softmax by choosing a large enough step-size.

C Other Proofs

C.1 Proof of Lemma 1

Proof. Recall the form of the cross-entropy loss for classification with K classes:

$$\ell(y, \xi) = -\sum_{k=1}^{N} \mathbb{1}\{y = k\} \log \frac{e^{\xi_k}}{\sum_j e^{\xi_j}}.$$

Its derivatives take the form

$$\frac{\partial \ell}{\partial \xi_k}(y, \xi) = s(\xi)_k - \mathbb{1}\{y = k\},$$

with $s(\xi)_k = \frac{e^{\xi_k}}{\sum_j e^{\xi_j}}$ the softmax.